THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Fish breathe through: 1

(a) lungs **(b)** gills **(c)** skin only **(d)** spiracles

2. Plants are called the producers because they: 1

(a) decompose waste **(b)** make food for all

(c) eat others **(d)** are green only

3. The image formed by a plane mirror is: 1

(a) real and inverted **(b)** virtual and erect

(c) real and erect **(d)** virtual and inverted

4. A solar eclipse occurs when the ___ comes between the Sun and Earth. 1

(a) Mars **(b)** Venus **(c)** Moon **(d)** Earth

5. The formula of sulphuric acid is: 1

(a) HNO_3 **(b)** $NaOH$ **(c)** HCl **(d)** H_2SO_4

6. Materials that do not allow electric current to pass through them are: 1

(a) conductors **(b)** metals **(c)** filaments **(d)** insulators

7. Which of the following is a non-metal in the gaseous state? 1

(a) mercury **(b)** sodium **(c)** oxygen **(d)** zinc

8. Which of the following is an irreversible change? 1

(a) folding paper **(b)** freezing water

(c) melting ice **(d)** ripening of fruit

9. The cracking of the voice in boys is due to growth of the: 1

(a) heart **(b)** voice box (larynx)

(c) lungs **(d)** stomach

10. In convection, heat is carried by: 1

(a) moving particles of the fluid **(b)** waves in space

(c) vibrating solid **(d)** light only

11. The SI unit of speed is: 1

(a) metre per second **(b)** kilometre **(c)** second **(d)** kilogram

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12. The process of breaking down complex food into simple substances is: 1

(a) respiration **(b)** digestion **(c)** egestion **(d)** absorption

13. The process by which green plants make food using sunlight is: 1

(a) digestion **(b)** respiration **(c)** photosynthesis **(d)** transpiration

14. Objects that give out their own light are: 1

(a) transparent objects **(b)** non-luminous objects

(c) opaque objects only **(d)** luminous objects

15. One rotation of the Earth takes about: 1

(a) 12 hours **(b)** 24 hours **(c)** 365 days **(d)** 30 days

16. Neutral substances: 1

(a) turn red litmus blue **(b)** turn blue litmus red

(c) do not change the colour of litmus **(d)** turn both red

17. The heating effect of electric current is used in: 1

(a) radio **(b)** electric iron **(c)** compass **(d)** fan only

18. Metal oxides are generally: 1

(a) sour **(b)** basic **(c)** acidic **(d)** neutral

19. Rust is chemically: 1

(a) iron carbonate **(b)** iron chloride

(c) hydrated iron oxide **(d)** iron sulphide

20. In general, the growth spurt starts earlier in: 1

(a) girls **(b)** boys

(c) both at exactly the same age **(d)** neither of them

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. How can you make a battery from cells? 2

22. Name a metal and a non-metal that are liquid at room temperature. 2

23. What are the conditions for rusting? 2

24. What is menstruation? 2

25. Why do we wear light-coloured clothes in summer? 2

26. A boy runs 400 m in 80 s. Find his speed. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Why are heaters placed near the floor and coolers/ACs near the ceiling? 3

28. Convert: 72 km/h to m/s and 15 m/s to km/h. 3

29. Explain how the heart pumps blood in the body. 3

30. Explain how a pitcher plant gets food. 3

31. A person stands 2 m in front of a plane mirror. Find the distance of the image from the mirror and between the person and the image. 3

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OR

How does a lunar eclipse occur? 3

32. Why does a tooth decay get worse with acid? 3

OR

What happens when two cells are connected in series with the same polarity? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Describe the human excretory system and the formation of urine. 5

34. Describe how plants respire and excrete. 5

35. Describe a kaleidoscope: how it is made and the pattern formed. 5

36. Describe the features of the Moon and compare with the Earth. 5

OR

Explain neutralisation with the example of an acid and a base. Write the equation and state two applications. 5

37. Explain the working of a simple electric circuit and the function of cell, bulb, switch and wires. 5

OR

Explain the physical properties of non-metals and compare them with metals in a table-like form. 5

38. Explain the process of rusting, its conditions and the methods of preventing it. 5

OR

Explain the endocrine glands, their secretions and functions with examples. 5

END OF QUESTION PAPER — Total: 80 Marks

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